

Yasir Arafat

Mechanical Engineering Student
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EDUCATION

BSc Mechanical Engineering
HAN University of Applied
Sciences
Arnhem, Netherlands
2024 – Present

CERTIFICATIONS

- Safety for Operational Supervisors VCA

CAD & SIMULATION

- SolidWorks CAD
- FEM Simulation
- Stress, strain & deformation
- Buckling & factor of safety
- Assembly & weldment design
- Technical drawings / GD&T

PROTOTYPING & MFG

- FDM 3D Printing (PLA)
- Cura slicer & support analysis
- Laser cutting
- Lathe & milling machining
- Mechanical assembly

OTHER

- Prototype testing & validation
- Technical reporting
- Team collaboration

LANGUAGES

English C1
Bangla Native

PROFILE

Mechanical Engineering student with hands-on experience in SolidWorks CAD, FEM simulation, and prototype manufacturing. Across three industry-connected projects I have worked with 3D printing, laser cutting, lathe machining, and iterative design for manufacturability. Comfortable taking ownership of tasks, working independently, and contributing to multidisciplinary teams.

EXPERIENCE

Project Engineer Sep 2025 – Jan 2026

Fydro Baking Stones

- Worked in a multidisciplinary project along with other engineering students at Fydro. The task was to come up with a solution for Fydro's yearly 500 tons of GFRC (Glass Fiber Reinforced Concrete) production residue and how it can be made into a new product in order to close the loop.

PROJECTS

CORDS VTS: Cyclist Safety Dummy System Feb 2026 – Jun 2026

- Designed articulated dummy components in SolidWorks with a focus on plastic component design and functional assembly.
- Optimised every part for 3D printability: adjusted print orientation, minimised overhangs and support material through Cura slicer analysis.
- Manufactured prototype parts using FDM 3D printing, laser-cut EVA foam, and PVC structural components with magnetic attachment systems.
- Iterated through rapid design-test-refine cycles to meet functional and reusability requirements.

Mechanical Battery: Planetary Gear Energy Storage Feb 2025 – Jun 2025

- Designed a planetary gear-based mechanical energy storage system in SolidWorks, including rotational assemblies and structural component layout.
- Manufactured components using lathe machining, FDM 3D printing (PLA springs), and laser-cut wooden housing.
- Tested, analysed, and documented system efficiency, achieving ~33% measured efficiency against a theoretical ~55%.

Case Project: Redesign of Industrial Vibrating Conveyor Structure Sep 2025 – Jun 2026

- Redesigned an industrial vibrating conveyor structure using SolidWorks CAD and FEM analysis to improve structural stability and reduce deflection.
- Performed static, frequency, and buckling simulations to evaluate stress distribution, vibration behaviour, and structural safety under operational loads.
- Validated simulation results and optimized the frame design for improved performance and durability.